

DFA & NFA PRACTICE SHEET

AUTOMATA & COMPUTABILITY (CSE331)

1. Starts With

1. Draw a DFA for the set of binary strings that start with 01 . $\Sigma = \{0,1\}$
2. Draw a DFA for the set of strings that start with a . $\Sigma = \{a,b\}$
3. Draw a DFA for the set of binary strings that start with 101 . $\Sigma = \{0,1\}$
4. Draw a DFA for the set of binary strings that start with odd number of 0s. $\Sigma = \{0,1\}$

2. Ends With

1. Draw a DFA for the set of strings that end with abb . $\Sigma = \{a, b, c\}$
2. Draw a DFA for the set of strings that end in 11 . $\Sigma = \{0,1\}$
3. Draw a DFA for the set of strings that end with 01 . $\Sigma = \{0,1\}$
4. Draw a DFA for the set of strings that don't end with 01 . $\Sigma = \{0,1\}$
5. Draw a DFA for the set of strings that don't end with 0101 . $\Sigma = \{0,1\}$

3. Contains

1. Draw a DFA for the set of strings that contain 011 as a substring and 001 as not a substring. $\Sigma = \{0,1\}$
2. Draw a DFA for the set of strings that contain 001 . $\Sigma = \{0,1\}$
3. Draw a DFA for the set of strings that have 011 as a subsequence and 001 as not a subsequence. $\Sigma = \{0,1\}$

4. Does Not Contain

1. Draw a DFA for the set of strings that does not contain the string $aabb$. $\Sigma = \{a,b\}$
2. Draw a DFA where the string does not contain consecutive '0s'? $\Sigma = \{0,1\}$

5. Length Constraints

1. Draw a DFA for the set of strings that have a length of at least 4 . $\Sigma = \{a, b\}$
2. Draw a DFA for the set of strings that have lengths of not more than 3 . $\Sigma = \{0,1\}$
3. Draw a DFA for the set of binary strings of even/odd length. $\Sigma = \{0,1\}$
4. Draw a DFA for the set of strings whose length is one more than a multiple of 3 . $\Sigma = \{0,1\}$
5. Draw a DFA for the set of strings of length exactly 3 . $\Sigma = \{0,1\}$
6. Draw a DFA for the set of strings that have lengths of less than 3 . $\Sigma = \{0,1\}$
7. Draw a DFA for the set of strings with a length of at least 2 . $\Sigma = \{0,1\}$

8. Binary strings which have a 0 at its third position.
9. Binary strings which have a 1 at its second last position.

6. Count Constraints

1. Draw a DFA for the set of binary strings that contain at least three 1s. $\Sigma = \{0,1\}$
2. Draw a DFA for the set of binary strings that contain at least three 1s consecutive. $\Sigma = \{0,1\}$
3. Draw a DFA for the set of strings that have exactly two 0s. $\Sigma = \{0,1\}$
4. Draw a DFA for the set of strings that have at most two 0s. $\Sigma = \{0,1\}$
5. Draw a DFA for the set of strings that have exactly three as. $\Sigma = \{a, b, c\}$
6. Draw a DFA for the set of strings where the number of 1s is not an integer multiple of 4. $\Sigma = \{0,1\}$

7. Character Patterns

1. Draw a DFA for the set of strings that have three consecutive 1s. $\Sigma = \{0,1\}$

8 Unique Type

1. Draw a DFA for the string w , when interpreted as a binary number is divisible by 3 .
2. Draw a DFA for the string w , when interpreted as a binary number divisible by 5 .
3. Binary strings whose decimal equivalent is divisible by 6.
4. Draw a DFA for the string w , when interpreted as a binary number divisible by 8 (don't use more than 4 states)
5. Draw a DFA for the set of strings where the first and last character are the same, and the length is greater than 1. $\Sigma = \{a,b\}$
6. Draw a DFA for the set of strings where the first and last character are the same, and the length is greater than 0. $\Sigma = \{a,b\}$

9. New Addition

1. Draw a DFA for the set of binary strings **every** 3rd position in w is 1. $\Sigma = \{0,1\}$
2. Draw a DFA where string contains '00' exactly once? $\Sigma = \{0,1\}$
3. Draw a DFA for the string w, starts and ends with different symbols, $\Sigma = \{0,1\}$
4. Draw a DFA for which 0's and 1's alternate in w.
5. Construct a DFA defined as $L = \{w \mid \text{each "b" is followed by at least one "a"}\}$ $\Sigma = \{a,b\}$ For example: baaa
6. Construct a DFA defined as $L = \{w \mid \text{each "b" is followed by at most one "a"}\}$ $\Sigma = \{a,b\}$
7. Draw a DFA where w starts with a^n , where n is a odd/even number (a^n means n consecutive a's)
8. Draw a DFA where the count of 'a' in w is not a multiple of 4.
9. $\{w \in \Sigma^* \mid \text{the sum of the symbols of w is a multiple of 3}\}$, where $\Sigma = \{0, 1, 2\}$.
10. $\{w \in \Sigma^* \mid w \text{ is any string not in } 0^*1^*\}$, where $\Sigma = \{0, 1\}$.
11. The set of binary numbers has 0 in all even positions. $\Sigma = \{0,1\}$.
12. Draw a DFA which accepts exactly two "00" as a substring.

13. Construct a DFA that accept the language, $L = \{ w \in \{a,b\}^* : w \text{ starts and ends}$

with different symbols.}

14. Construct a DFA that accept the language, $L = \{ w \in \{a,b\}^* : w \text{ starts and ends}$

with the same symbol.}

15. $\{w \mid w \in \{a, b, c\}^* w = a^m b^n c^l \text{ for } m, n, l \geq 1\}$

16. where 0's and 1's appear in alternating groups of odd numbered length.

17. $\{w \mid w \in \{0, 1\}^* w \text{ contains an equal number of occurrences of the substrings } 01 \text{ and } 10 \}$

For any DFA there Exists an NFA as well. Try to Draw an NFA as well.

Regular Operations on DFA:

1. Draw a DFA for a string that contains at least one 1 and an even number of 0s follows the last 1.

2. Draw a DFA for binary strings that have an even number of 0s or an odd number of 1s. $\Sigma = \{0,1\}$

3. Strings that contain 011 as a substring and 001 as not a substring. $\Sigma = \{0,1\}$

4. Strings that do not contain "ba" and end with "cb". $\Sigma = \{a,b,c\}$

5. $L_1(M) \rightarrow \{w \in \Sigma^* \mid w \text{ doesn't contain } 00\}$, where $\Sigma = \{0, 1\}$. $L_2(M) \rightarrow \{w \in \Sigma^* \mid w \text{ doesn't contain } 11\}$, where $\Sigma = \{0, 1\}$. ,
find $L(M) \rightarrow (L_1 \cap L_2)$.

Regular Operations on NFA:

1. Design an NFA which contains at least two 0's or exactly f 1's.
2. Design an NFA which accepts an even number of 0's or number of 0's which are divided by 3.
3. Design an NFA which contains "000" or "111".
4. Design an NFA where all strings contain exactly six 0s or an odd number of 1s.
5. Design an NFA where the set of strings begin or end (or both) with 01
6. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{length of } w \text{ is a multiple of } 2 \text{ or } 3\}$, where $\Sigma = \{0, 1\}$.

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Editing in progress

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